NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

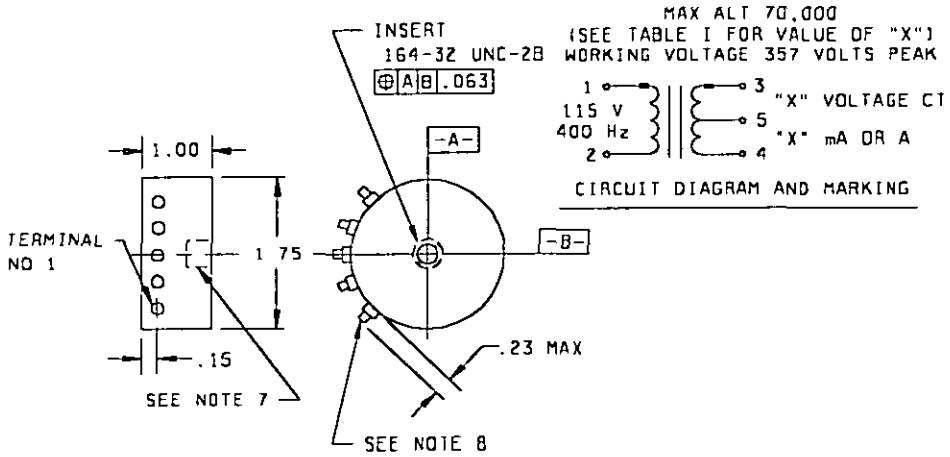
MIL-PRF-27/351D
22 January 1992
SUPERSEDING
MIL-T-27/351C
29 March 1989

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 25 VOLT-AMPERES, 400 HZ, TF5S03ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.



Inches	mm
.063	1.60
.15	3.8
.164	4.17
.23	5.8
1.00	25.4
1.75	44.5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. Dimensional tolerance shall be ±.03 (0.8 mm) unless otherwise specified.
4. Marking shall be on the top case.
5. Terminals shall be identified by numbers 1 through 5 marked on the top of case above each terminal.
6. Secondary electrical values shall be marked as specified in table I, as applicable.
7. The insert shall be 5 full threads minimum.
8. Terminal spacing is .35 ±.03 (8.9 ±0.8 mm).

FIGURE 1. Dimensions and configuration.

(D) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS: (When numbers in parentheses i e (1-2) are used, they indicate the winding and the extreme terminals of the winding)

Electrical ratings See table I.

Primary voltage (1-2) 115 volts ±10 percent, 400 hertz ±5 percent

Working voltage: 357 volts, peak

Design and construction

Dimensions and configuration See figure 1

Weight 33 pound, maximum

Duty cycle Continuous

Case Epoxy fiberglass

Altitude 70,000 feet, maximum

Operating temperature range -55°C to +130°C

Terminals Feedthrough

Terminal height 0 23 inch, maximum.

Terminal strength MIL-STD-202, method 211, test condition A, 5 pounds

Dielectric withstanding voltage

At sea level 1000 volts, rms

At reduced barometric pressure 1 25 times the peak working voltage specified

Electrical characteristics

No load With 115 volts, 400 hertz across (1-2), current in (1-2) shall not exceed 25 milliamperes rms

Rated load With 115 volts, 400 hertz across (1-2) and a resistive load with current as specified in table I in (3-4), the voltage across (3-4) shall be as specified in table I.

Efficiency 90 percent minimum at rated load

Polarity Additive with terminals 2 and 3 connected

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$$\text{Regulation} \quad \frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100 \quad \text{(Shall not exceed 10 percent.)}$$

Temperature rise. 40°C maximum with 115 volts, 400 hertz across (1-2) at an ambient temperature of 90°C.

Marking location See figure 1

Part or Identifying Number (PIN) example

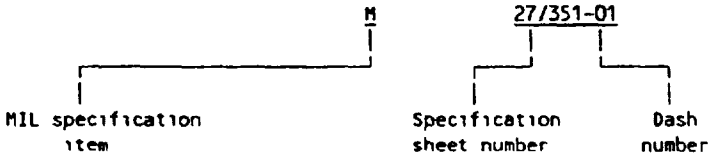


TABLE I Electrical ratings

Dash number M27/351	Secondary voltage 1/ (3-4) (V rms) ±5%	Secondary current (3-4) (amps rms)
-01	5 0	5.00
-02	7 0	3 57
-03	10.0	2 50
-04	12 0	2.08
-05	14 0	1 79
-06	17 0	1 47
-07	24 0	1 04
-08	28 0	0 89
-09	34 0	0 74
-10	115.0	0 21
(D) -11	26 0	0 890

1/ One-half of the listed output voltage is available between pins 3 and 5 or 4 and 5

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QUALITY ASSURANCE PROVISIONS

Extent of qualification

(D) Qualification testing and approval to M27/350-11 and M27/354-09 shall be sufficient to grant qualification approval to MIL-T-27/350 through MIL-T-354 inclusive, all parts

(D) Qualification testing and approval to M27/351-11 shall be sufficient to grant qualification approval to M27/351-01 through M27/351-11

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR, ME, WC
Navy - AS, MC
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0780)

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